CLOUD COMPUTING

Course Introduction

- Acknowledgements: this material draws on the course's prescribed textbooks, prior year lecture notes prepared by the course staff, and publicly available reference material from equipment and software vendors, adapted here for instructional use only and not intended for redistribution outside this course.

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Hypervisors

- A hypervisor abstracts physical hardware for virtual machines.
- Type 1 hypervisors run directly on hardware; Type 2 run atop a host OS.

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Virtualization Techniques

- Full virtualization traps and emulates privileged instructions.
- Paravirtualization requires a modified guest OS for better performance.

Memory Virtualization

- Shadow page tables track guest-to-host physical address mappings.
- This technique reduces virtualization overhead significantly.

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Popek-Goldberg Theorem

- Sensitive instructions must be a subset of privileged instructions for efficient virtualization.

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VM Migration

- Live migration moves a running VM with minimal downtime.
- Cold migration requires stopping the VM before moving it.

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Containers

- Containers share the host kernel instead of virtualizing hardware.
- This technique reduces virtualization overhead significantly.